EXPERIMENT - 63

1. Problem Statement

Write a C program that reads the size and elements of a variable-length integer array, nums[]. The array contains distinct integers from 0 to size, with exactly one number missing. Pass the array to a function named missing_number() using its address. Use pointers to pass the array between the calling function and missing_number(). The function missing_number() must:

- Find the missing integer from the range 0 to size.
- Display the missing number.
- Assume that all elements in the array are distinct.
- Use the fact that the array contains exactly n numbers from the range 0 to size, with one number missing.
- Display the result as an integer.

2. Algorithm

Main Program

Step 1: START
Step 2: Read n and declare the variable length array nums of size n.
Step 3: Read the n elements of nums.
Step 4: Call missing_number(&nums[0], n).
Step 5: STOP

Function missing_number()

Step 1: Set sum = 0 and add every element of nums to sum.
Step 2: Find the total of 0 to n as n * (n + 1) / 2.
Step 3: Display total − sum, which is the missing number.

3. Test Case Table

#	Size	Array	Expected Output	Actual Output	Result
1	3	3 0 1	Missing number: 2	Missing number: 2	**PASS**
2	2	0 1	Missing number: 2	Missing number: 2	**PASS**
3	9	9 6 4 2 3 5 7 0 1	Missing number: 8	Missing number: 8	**PASS**
4	1	0	Missing number: 1	Missing number: 1	**PASS**
5	3	1 2 3	Missing number: 0, 0 itself is missing	Missing number: 0	**PASS**
6	5	0 1 3 4 5	Missing number: 2	Missing number: 2	**PASS**

4. Source Code

missing63.c

```
/* Experiment 63: find the missing number from 0 to n using the sum. */
#include <stdio.h>

void missing_number(int *nums, int n);

int main(void)
{
    int n, i;

    printf("Enter the size of the array: ");
    scanf("%d", &n);
    int nums[n];
    printf("Enter %d elements: ", n);
    for (i = 0; i < n; i++)
        scanf("%d", &nums[i]);

    missing_number(&nums[0], n);
    return 0;
}

/* The numbers 0 to n add up to n*(n+1)/2. Whatever the array
   is short of that total is the missing number. */
void missing_number(int *nums, int n)
{
    int i, sum = 0;

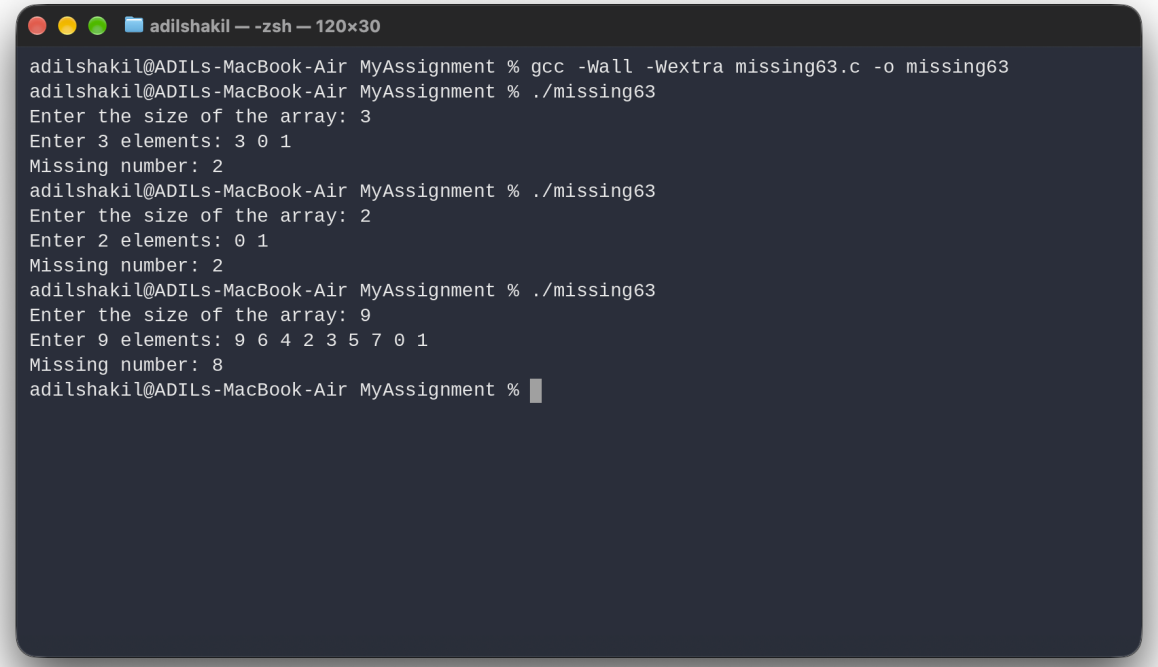
    for (i = 0; i < n; i++)
        sum += *(nums + i);

    printf("Missing number: %d\n", n * (n + 1) / 2 - sum);
}
```

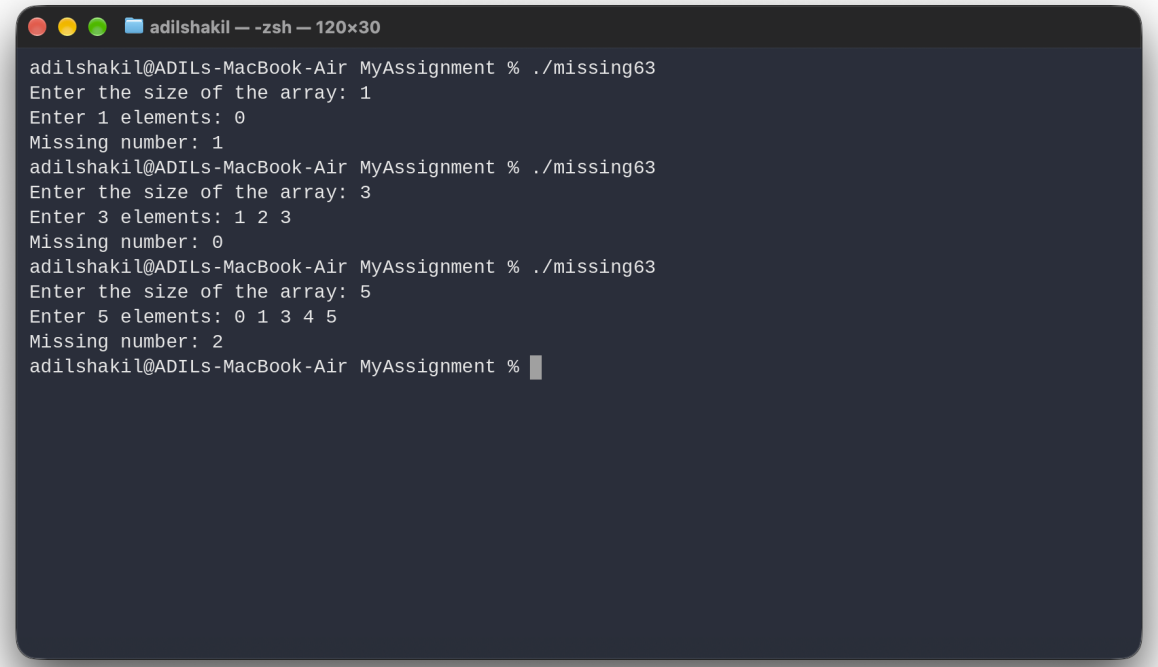
5. Compilation and Output

Compiled with `gcc -Wall -Wextra missing63.c -o missing63` so the small warnings would show up too. It built clean.

Cases 1 to 3



Cases 4 to 6



All six passed, and the first four are the examples from the question.